PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Training and Redeployment of Healthcare Workers to Intensive Care Units (ICUs) During the COVID-19 Pandemic: a Systematic Review
AUTHORS	Vera San Juan, Norha; Clark, Sigrún; Camilleri, Matthew; Jeans, John; Monkhouse, Alexandra; Chisnall, Georgia; Vindrola-Padros, Cecilia

VERSION 1 – REVIEW

REVIEWER	Szabo, Rebecca The University of Melbourne, Obstetrics and Gynaecology
REVIEW RETURNED	28-Mar-2021

GENERAL COMMENTS	Thank you for this work. A timely addition to the literature. Overall well written and clear but there are a few areas that could be clearer. Mostly the authors refer to ICU but in a few locations reference emergency services. Consistency with either ICU or critical care would be helpful. A few articles in this area seem to be missing and I'm struggling to understand why based on your methods - namely this article from Bohmer et al NEJM 2020 on the Nightingale Learning Systems. It would be useful to understand why and if this was reviewed then not included or missed in the search. There are a few others also one of which have been published 2021 so understandably outside your search. Several places the pandemic is referred to in the past tense. As it is ongoing I'd suggest ensuring it is present tense throughout. Additionally in the conclusion and end of discussion you could outline your "so what" more clearly regarding relevance to this pandemic and/or future pandemics. That seems a bit underdone currently. My only other comment is around review of papers in other languages. I'm surprised China, Italy, Spain are not more represented given you had no English language limitation. This would be interesting to comment on in the discussion given it was not a limitation and given the extent of redeployment seen in those countries early in the pandemic. In methods I'm also missing how non-English papers were screened, read, translated. This is important to add. One sentence makes no sense, I think because of a typo, missing word or I'm missing something. The first line page 16, not clear what this sentence means, please clarify.

	There are several areas the word count could be reduced and be more concise. Also in patient and public section - did any of the researchers have covid19 or know of family or friends with covid? If yes they could be considered consumers so this is something to consider given the large number of HCWs who have been impacted personally during the pandemic. I am pleased to see plan for wide dissemination. A visual representation of the themes, as described for dissemination, might be useful to see with the paper. I wonder if the consumer here is in fact health services and HCWs rather than other groups...some food for thought.

REVIEWER	Meyer , Lois University of New South Wales, Medicine
REVIEW RETURNED	05-Apr-2021

GENERAL COMMENTS	This is a very well conceived, timely and relevant piece of research. The clear scope and focus of the study on redeployment and training in ICU settings during COVID-19 within a limited time period and lessons to be learnt going forward is an original and worthwhile study. The systematic review methodology is clearly laid out and summary results fairly well described. Where the article could be enhanced is in the quality of expression and clarity of presentation on the results. I also note that 'the principles of redeployment' appears to be misplaced as something to be found within the studies, rather than something distilled by you from the systematic review. I suggest owning this as a key outcome from the study and highlighting this in the Discussion to inform future practice as the three principles are very useful and align to your study's overall purpose. Please find the detailed review in the pdf attached. study on redeployment and training in ICU settings during COVID-19 and lessons to be learnt is an original and worthwhile study. The systematic review methodology is well laid out and the summary results are fairly-well described. Where the article could be a little enhanced is in the quality of expression and clarity of presentation on the thematic synthesis of results. The Discussion and Conclusion could more tightly demonstrate the overall value of the findings emphasising the value of the key principles that can inform future practice for redeployment and training in ICU settings within the evolving context of COVID-19. There are two areas where the paper could be improved conceptually. First is in relation to the first research question: What were the main strategies developed to redeploy staff to ICU? The second is in relation to the second research question: What were the principles of redeployment? Both questions are stated in the paper as guiding the systematic review. The issues attached to each are noted below. In relation to the first research question the paper does present a range of findings on strategies that were used to deploy staff including those that were more successful. What would have been useful would have been to accompany this finding with some brief discussion of who and how deployment was undertaken within an organisational/health system context. While the findings present processes such as the need for clear decision making there is a

	total absence of information on who is involved in undertaking this critical work and the implications for the success or otherwise of the strategies. For example on p. 13 it would be informative to know from the systematic review what was found in who were making the decisions and was this important to what occurred and how in different contexts? If this information was missing from the systematic review findings it would be worth noting and if they were present, the role of leadership and decision-making would be important to identify. The paper will be enhanced by more clearly including some brief discussion of the role of health leadership and decision makers in the redeployment process in different healthcare settings and systems as part of how redeployment and training occurs. This issue can have significant implications for understanding and interpreting future redeployment planning within a health systems perspective. In relation to the second research question the paper presents 'the principles of redeployment' as if these will be something that will be identified within the practices of the studies included in the systematic review. However, it is apparent that the three principles identified from the systematic review are the outcome of the distilled thematic analysis by the authors of this paper rather than agreed principles guiding real-world practice within the healthcare settings from the research papers themselves within the review. It is recommended the paper is amended so that it is clear the principles were not something informing practices found in the studies but rather are an outcome from the systematic review. It is recommended the authors own this as a key outcome from the study and highlight this in the Discussion for informing future practice as the three principles are significant and align to the study's overall purpose. It is also noted that in enhancing the paper reference should be made to the limitations of the paper occurring within the first 12 months of COVID-19 given the changing nature of the pandemic; ie the implications of the findings may need to be further interpreted and adapted. This point is suggested within the body of the paper especially given the three principles and the need for flexibility but it could be more clearly made. It is suggested that in the Conclusion not only the originality of the study is noted but also its timing so that the importance of the study's findings are reinforced within the continuing evolving and global context of COVID-19. Suggested changes in expression and punctuation: Abstract: Line 7-8 Remove the word 'Reluctantly' Change the sentence to 'This has required the… Introduction: Line 32-33. Include reference for the assertion that "The impact of redeployment…has also been identified". Synthesis Findings Theme 1: Redeployment: Implementation strategies and learnings: Paragraph one lines 12 – 15 provide little more than in Table 3 and need clearer explanation. This lack of discussion is important as goes to the research question on what were the main strategies used to deploy health care staff? Did different facilities use different methods? Did it change across the year? What were the considerations? Who were making the decisions and was this important to what occurred and how in different contexts? These more systemic issues are missing and if so from the papers in the systematic review it would be worth noting and if they were present important to future redeployment planning.

	Theme 3. Redeployed staff learning needs: Line 21 Need to change tense from will be to "……. work experience and focused solely on content that was relevant for redeployed roles" Theme 3. Redeployed staff learning needs: Second paragraph, lines 26 -29 It is unclear if the training needs discussed was found across all papers in the systematic review or was this something identified in only some. The paragraph also needs rewriting as currently poorly expressed as series of points. Theme 3. Redeployed staff learning needs: Last paragraph lines 38 – 40 This one sentence paragraph needs expansion into further explanation as paragraphs should be more then one sentence. Also this is insufficient detail and is an important point as it is referred to later in the Discussion. Theme 4. Training formats offered and training evaluations: Training formats offered and training evaluations:p.15 Lines 16 -20 These following lines need clarification. "However, innovations in the peer review process and format of presenting results helped facilitate the continuation of this mode of sharing experiences [8]. Despite these efforts, reaching a single message for best practice was a significant challenge during the pandemic and affected training development" The first sentence here is unclear – what exactly is meant here after stating sharing peer review papers was not effective in the pandemic context. Also. what is meant by reaching a single message affected training development? It is as if all the health institutions in the systematic review were aware of what was to be known and were sharing the information at the time. Theme 5. Future redeployment and training concerns: p. 15 Lines Lines 36 -37. This needs to be rephrased as a finding from the systematic review as a whole not as if this was something occurring across all institutions unless that was the case. There is a blurring of positioning in the writing. Please refer to Theme 2 which is the best presented of the themes in clearly showing what was found from the systematic review. Theme 5. Future redeployment and training concerns: p. 15 Lines Lines 44-46. This paragraph needs incorporation more fully within the section on shifting needs. This section isn't really about Training Concerns...Did the systematic review reveal training concerns in considering the future or simply the need to consider flexible approaches for changing contexts? Suggest possibility of changing the title for this Theme and revising the writing of paragraphs here slightly to more clearly highlight the theme.

VERSION 1 – AUTHOR RESPONSE

REVIEWER 1

 Thank you for this work. A timely addition to the literature. Overall well written and clear but there are a few areas that could be clearer. Mostly the authors refer to ICU but in a few locations reference emergency services. Consistency with either ICU or critical care would be helpful.

Thank you for highlighting the timeliness of our work. We have changed the terminology to be consistent throughout the manuscript.

- A few articles in this area seem to be missing and I'm struggling to understand why based on your methods - namely this article from Bohmer et al NEJM 2020 on the Nightingale Learning Systems. It would be useful to understand why and if this was reviewed then not included or missed in the search. There are a few others also one of which have been published 2021 so understandably outside your search.

We decided it was best to update the search, and found an additional nineteen papers. We also added the article that you recommended, thank you. It is possible that it had not come up in our initial search because it did not include the word “redeployment” in the title or abstract.

- Several places the pandemic is referred to in the past tense. As it is ongoing I'd suggest ensuring it is present tense throughout.

We changed this, thank you.

- Additionally in the conclusion and end of discussion you could outline your "so what" more clearly regarding relevance to this pandemic and/or future pandemics. That seems a bit underdone currently.

We have added more detail around the possible applicability of findings in the conclusion.

- I'm surprised China, Italy, Spain are not more represented given you had no English language limitation. This would be interesting to comment on in the discussion given it was not a limitation and given the extent of redeployment seen in those countries early in the pandemic. In methods I'm also missing how non-English papers were screened, read, translated.

We have added a line in the limitations to reflect on this: “It is also possible that some studies from non-English speaking countries were not identified due to all the search terms being in English.” We also added a line to the methods to describe this: “In the event that non-English papers were identified, members of the broader team who spoke the language were asked to review it.”

- The first line page 16, not clear what this sentence means, please clarify. There are several areas the word count could be reduced and be more concise.

Unfortunately, page 16 in our version of the document is part of the reference list, so we could not identify the line that you mention. However, we have reviewed the manuscript and hope we have addressed your comment.

- Also in patient and public section - did any of the researchers have covid19 or know of family or friends with covid? If yes they could be considered consumers so this is something to consider given the large number of HCWs who have been impacted personally during the pandemic.

We have added a sentence to reflect our personal experience with COVID-19.

- A visual representation of the themes, as described for dissemination, might be useful to see with the paper.

We added the infographic outlining our findings to the supplementary material.

REVIEWER 2

- This is a very well conceived, timely and relevant piece of research. The clear scope and focus of the study on redeployment and training in ICU settings during COVID-19 within a limited time

period and lessons to be learnt going forward is an original and worthwhile study. The systematic review methodology is clearly laid out and summary results fairly well described.

Thank you very much for your positive feedback.

 I note that 'the principles of redeployment' appears to be misplaced as something to be found within the studies, rather than something distilled by you from the systematic review. I suggest owning this as a key outcome from the study and highlighting this in the Discussion to inform future practice as the three principles are very useful and align to your study's overall purpose.

Thank you for your suggestion and for highlighting the importance of our work. We have clarified this in the results and highlighted it in the discussion.

 There are two areas where the paper could be improved conceptually. First is in relation to the first research question: What were the main strategies developed to redeploy staff to ICU? The second is in relation to the second research question: What were the principles of redeployment?

Thank you for your thorough reflection, we kept your suggestions present in our edits throughout the document. Within the constraints of the word limit, we included more details around decision-making processes and the role of health leadership in the results and discussion; and we highlighted the originality of the study and implications in the conclusion.

 Suggested changes in expression and punctuation.

We added all the suggested expression changes and addressed your recommendations regarding the content within the constraints of the word limit.

VERSION 2 – REVIEW

REVIEWER	Szabo, Rebecca The University of Melbourne, Obstetrics and Gynaecology
REVIEW RETURNED	06-Dec-2021

GENERAL COMMENTS	Well done on this excellent addition to the literature.